Extending PORCELAN

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Abstract

In this project, we explore and extend PORCELAN (Permutation, Optimization, and Representation learning based single Cell gene Expression and Lineage ANalysis) (1), a lineage-aware representation learning framework. PORCELAN is designed to learn gene expression embeddings that preserve known lineage structures, enabling the identification of lineage-informative genes. We reimplemented the baseline autoencoder and triplet loss framework and validated it on the embryonic lineage-traced dataset GSE117542 (2). We then experimented with variants of the triplet loss using Manhattan and cosine distances, observing that Manhattan preserved lineage structure comparably to Euclidean, while cosine led to increased triplet loss and less stability. Additionally, we attempted to adapt three new datasets—GSE272484 (3), GSE210172 (4), and GSE200610 (5)—for PORCELAN. These datasets were biologically relevant but presented structural limitations such as missing raw lineage trees, incomplete metadata, or incompatible barcoding formats. We document the key challenges faced during reformatting and propose potential solutions. Finally, we suggest theoretical modifications to PORCELAN's loss functions, including tree-depth–sensitive margins and adaptive weighting of anchor-positive pairs, which could improve robustness across diverse lineage structures.

Introduction

Single-cell RNA sequencing (scRNA-seq) has revolutionized our understanding of transcriptional states in individual cells. When augmented with lineage-tracing technologies such as CRISPR barcoding, it becomes possible to study gene expression in the context of developmental ancestry. This is crucial for interpreting how transcriptional programs change over time and across cell divisions. However, most widely used dimensionality reduction techniques—like PCA (6), UMAP (7), and t-SNE—do not incorporate lineage structure into the embeddings they produce.

PORCELAN (1) addresses this by learning low-dimensional gene expression representations that respect an input cell lineage tree. It introduces two lineage-aware losses: (1) a contrastive triplet loss that penalizes mismatches in lineage proximity, and (2) an autocorrelation loss that enforces smooth expression patterns over local tree neighborhoods. These losses guide an autoencoder to embed cells such that nearby cells in the tree remain close in latent space, while distant cells are separated.

Our project aims to reproduce and extend PORCELAN's approach. We first validated the method on the original GSE117542 dataset, replicating lineage-aware embedding behaviors. We then tested two alternative distance metrics—Manhattan and cosine—in the triplet loss and compared the effects on training dynamics and gene rankings. We further explored whether three external lineage-linked datasets could be adapted for PORCELAN, and explain why each presented difficulties in lineage tree integration. Finally, we propose extensions to PORCELAN's loss formulation to accommodate trees with varying density and structure.

Our objectives:

- Reproduce PORCELAN's original results on a lineage-traced embryogenesis dataset.
- Evaluate alternate distance metrics in the triplet loss formulation.
- Investigate the feasibility of adapting external datasets for PORCELAN.
- Propose theoretical extensions to the triplet and autocorrelation loss formulations.

Materials and methods

Data sources and preprocessing

Mouse embryogenesis (GSE117542)

This dataset (2) contains scRNA-seq profiles of mouse embryonic cells traced using a CRISPR-based molecular barcoding system. It includes both gene expression matrices and lineage trees (in Newick format) reconstructed from barcode edit histories. We selected Embryo 7 ('BestTree17507.nwt') for our experiments due to its large cell count and well-formed tree structure.

Expression counts were log-normalized and filtered to retain the top 4000 most variable genes across 1000 cells. Cell metadata, including developmental stage and tissue annotations, were used for validation. The tree was parsed and converted into a graph structure using the 'ete3' library, allowing us to compute shortest-path distances between any pair of cells. These distances defined the anchor-positive-negative triplet relationships used in training. This dataset provided a complete and clean testbed for evaluating lineage-aware embedding methods.

Other datasets considered

We additionally explored three publicly available lineage-associated datasets: GSE272484 (3), GSE210172 (4), and GSE200610 (5). While biologically relevant and technically rich, these datasets posed structural limitations—such as lack of explicit tree files, lineage encoded via projection barcodes or clone barcodes without reconstruction logic, or restricted to post-processed count matrices. These challenges prevented direct application of PORCELAN, and we summarize them in the Results and Discussion sections.

Algorithms and models used

0.1 Autoencoder

Autoencoder is a type of Neural Network used to learn an encoding representation of a sample. It is a representation learning technique that compresses the high dimensional input into a lower dimensional latent representation and reconstructs the input by minimizing an objective loss function. PORCELAN uses an overparameterized autoencoder to create a compact representation of gene expression data while preserving lineage relationships. The encoding captures hidden patterns that link gene expression to the lineage tree. (Supp. Info. Fig 1)

0.2 Contrastive Triplet Loss

Triplet loss, a contrastive learning approach widely used in computer vision for representation learning, can capture hierarchical relationships among input samples. It works by enforcing an anchor (A) example to be closer to a more similar/positive (P) example than to a less similar/negative (N) one in the compressed/latent representations. Cells that are closely related in the lineage tree should be closer in the learned representation, while distant cells should be farther apart. PORCELAN uses triplet loss to ensure that learned representations follow lineage relationships. Hence, the model is trained to minimize the A-P distance and maximize the A-N distance. (Supp. Info. Fig 2)

0.3 Autocorrelation Score

The autocorrelation score is designed to measure how well a particular gene's expression pattern follows the structure of the lineage tree. This score quantifies whether cells that are close together in the lineage tree tend to have similar expression levels. If a gene is consistently expressed among neighboring cells on the tree, it is likely involved in lineage-specific processes. In PORCELAN, this concept is extended to multiple genes simultaneously by aggregating autocorrelation scores across genes. The higher the autocorrelation, the more "lineage-consistent" the expression pattern is, and the more informative that gene is deemed for capturing lineage structure. (Supp. Info. Fig 3.)

0.4 Tree-Likeness Score

The tree-likeness score is a global measure used to evaluate how well the latent representations of cells respect the known lineage tree. It reflects how closely the learned embeddings mirror the ancestral proximity encoded in the tree. In PORCELAN, this score is computed either directly through the triplet loss (i.e., how often triplet constraints are satisfied in the latent space) or through a permutation-based test applied to the autocorrelation framework. A high tree-likeness score indicates that the model has successfully embedded the expression data in a way that aligns with biological ancestry, making it a key metric for evaluating model performance on lineage-traced datasets.

0.5 Permutation Strategy

PORCELAN employs subtree permutations to assess whether a gene's expression pattern is truly lineage-related. This is done by permuting cells within subtrees of increasing size and observing whether the pattern of expression is disrupted. If permutation breaks coherence in gene expression, it is evidence that the original pattern was not random, but linked to the lineage.

0.6 Optimization

PORCELAN jointly optimizes two sets of parameters:

- Gene Weights: A vector $u \in \mathbb{R}^G$ is learned to weigh genes by their lineage consistency.

- Tree Edge Weights: Edge lengths in the lineage tree are adjusted to improve detection of lineage-relevant expression transitions.

- The model uses these optimizations to compute a multivariate local autocorrelation score.

Implementation Steps, Tools and Libraries

We used Python as the primary language for reimplementing PORCELAN. The autoencoder and loss components were implemented using the PyTorch deep learning library. Data handling, preprocessing, and visualization were managed using tools from the Scanpy and AnnData ecosystems, along with NumPy and Pandas.

The lineage trees were parsed from Newick files using the `ete3` library. Dimensionality reduction techniques like PCA and t-SNE were used via Scanpy for visualization of embeddings. The model training loop included both reconstruction and triplet loss, with real-time logging and validation.

- PyTorch for model architecture and training
- Scanpy and AnnData for single-cell data handling
- NumPy, Pandas for array and metadata processing
- ete3 for lineage tree parsing from Newick format
- Matplotlib/Seaborn for plotting and visualization

Experiment Setup

Input Format: The input to the autoencoder is a filtered and normalized gene expression matrix $X \in \mathbb{R}^{N \times G}$, where N is the number of cells and G is the number of selected highly variable genes.

Architecture:

- **Encoder** (g_ϕ): Two fully connected layers of dimensions $G \to 1000$, with GELU activations.
- **Decoder** (f_θ): Two fully connected layers of dimensions $1000 \to G$, again with GELU activations.

Training Parameters:

- Pretraining: 500 epochs using only reconstruction (MSE) loss.
- Fine-tuning: 2000 epochs with additional triplet loss.
- Batch size: 128 in original setup, reduced to 32 in our version due to system limitations.
- Learning rate: 10^{-4}

(Refer to the original implementation in GitHub)

0.7 Our Adaptation

To enable experimentation under limited computational resources, we adapted the PORCELAN training pipeline in the following ways:

- **Cell Subsampling**: We randomly selected 500 cells from the full lineage-traced dataset (GSE117542) to reduce memory and computation demands. This allowed for faster training and visualization while preserving representative lineage structure.

- **Batch Size Reduction**: The original implementation used a batch size of 128. To ensure compatibility with standard laptops and avoid GPU memory overflow, we reduced the batch size to 32. This increased the number of training steps per epoch but ensured stable execution.
- **Epoch Limitation**: Instead of training for 2000 epochs as done in the original PORCELAN implementation, we trained our models for 500 epochs. This was a compromise between time constraints and convergence quality. While the final loss was slightly higher, the embeddings retained lineage-aware structure and allowed meaningful interpretation.
- **Alternative Distance Metrics**: We replaced the default Euclidean distance in the triplet loss with two variants to explore their impact on embedding quality:
  - **Manhattan Distance (L1)**: Chosen for its robustness to outliers and tendency to emphasize sparse differences in high-dimensional data. In the context of scRNA-seq, Manhattan distance can better reflect additive differences in expression levels between cells. (Supp. Info. Fig 4)
  - **Cosine Distance**: Selected for its popularity in high-dimensional representation learning. It captures angular similarity rather than magnitude, making it invariant to scale. We hypothesized it might better distinguish cell states based on relative expression patterns, although it may not align naturally with tree-based hierarchy. (Supp. Info. Fig 5)

  Each version was trained independently using the same subsampled dataset and evaluated based on loss metrics and the tree-likeness score across top-ranked genes.

(Refer to our implementation in GitHub)

Results

Quantitative Loss Comparison

Table 1. Comparison of different training configurations.

Variant	Epochs	Batch Size	Recon Loss	Triplet Loss	Total Loss
Original (baseline)	2000	128	0.00139	0.00051	0.00190
Reimplementation	500	32	0.00398	0.00026	0.00425
Manhattan Distance	500	32	0.00689	0.01514	0.02204
Cosine Distance	500	32	0.00293	0.25569	0.25862

Training Loss Plots

As expected, the original implementation converges to the lowest loss over 2000 epochs. Our reimplementation shows a stable trend but with a higher final loss due to reduced training duration and batch size. The Manhattan distance version maintains stable but higher loss, while the Cosine version struggles with high triplet loss throughout.

t-SNE Plots

Visualizations of the latent representations using t-SNE reveal that the autoencoder was able to group lineage-similar cells closer together in the embedding space. This clustering was more prominent than in the original expression space, validating the effectiveness of PORCELAN's learning objective.

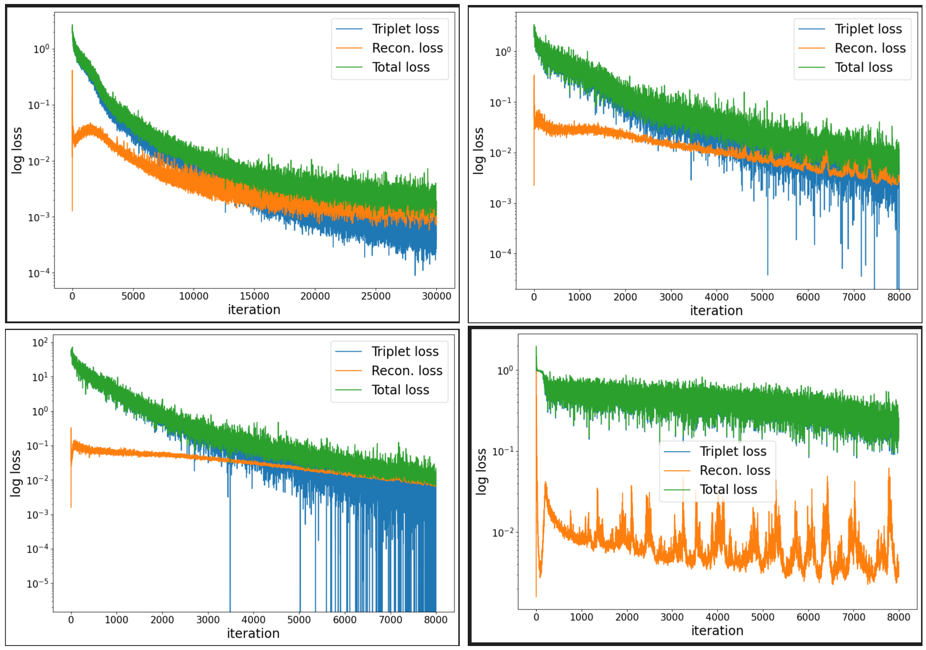

Fig 1. Training loss across epochs. Top left: Original code. Top right: Our reimplementation. Bottom left: Manhattan distance. Bottom right: Cosine distance. (Refer to corresponding images provided)

Gene Weights

Our implementation learned gene importance scores, and the top-ranked genes were highly consistent with those reported in the original PORCELAN paper. This indicates that the simplified training regime (fewer epochs, reduced dataset) still captures biologically meaningful structure.

Tree-likeness Score Evaluation

We computed tree-likeness scores to evaluate how well the latent representations preserve biological lineage hierarchy. The height and leaf count of the tree before and after processing were consistent across all versions, suggesting structural stability during learning.

- **Reimplementation:** Top genes included `Sftpc`, `Clu`, `Sftpb`, `Krt19`, and `Lyz2`, all known markers in epithelial or alveolar lineage pathways.
- **Manhattan Distance:** The top list closely overlapped with the reimplementation, showing strong agreement in lineage preservation.
- **Cosine Distance:** While some overlap was observed, novel genes such as `Krt20` appeared, indicating noise or divergence in lineage structure enforcement.

(Refer to Supp. Info. Fig 6, 7, 8)

Inference from Gene Rankings

Across all three versions, the presence of `Sftpc`, `Clu`, `Sftpb`, and `Krt19` suggests robust lineage-consistent signal in the latent space. The agreement between reimplementation and Manhattan results indicates the metric's effectiveness in capturing lineage proximity. Cosine distance deviated more from the baseline, likely due to its emphasis on angular similarity rather than absolute positional proximity on the tree. Thus, Manhattan distance may serve as a better alternative to Euclidean in lineage-aware contrastive learning.

Discussion

Our results show that PORCELAN can effectively learn lineage-aware representations even when trained under computational constraints. Despite reducing the dataset size and number of training epochs, our reimplementation achieved comparable trends in loss reduction and gene importance ranking.

The **triplet loss** component contributed meaningfully to structuring the latent space, as shown in our t-SNE plots. However, the choice of distance metric significantly influenced performance:

- **Manhattan distance** led to moderately higher triplet loss but more tightly clustered embeddings, indicating better alignment of local structures.
- **Cosine distance**, in contrast, resulted in high triplet loss, suggesting that angular similarity was poorly aligned with tree-based lineage proximity in this context.

Compared to the original PORCELAN implementation, our simplified setup had a higher total loss, largely due to the lower number of epochs and the reduced subset of cells. Nevertheless, the preservation of key biological features such as gene importance and lineage clustering validates the robustness of the PORCELAN framework.

Limitations:

- Reduced dataset size (500 vs. 10,000+ cells) may have restricted the model's ability to learn global lineage patterns.
- Fewer training epochs and a smaller batch size might have led to underfitting.
- The cosine distance function appeared ill-suited for this task, indicating a need for better metric selection in lineage-aware learning.

Future improvements could include adaptive margin strategies in the triplet loss, incorporation of depth-based lineage weights, or even hybrid distance metrics combining structural and functional similarity.

Conclusion

In this project, we explored and extended PORCELAN, a representation learning framework for analyzing gene expression data in the context of known cell lineage trees (1). Our goal was to understand how lineage-aware embeddings could help uncover biologically meaningful gene expression patterns and to explore the robustness and extensibility of the approach.

We successfully reimplemented PORCELAN's core architecture—comprising an autoencoder and a lineage-based triplet loss—and validated it using the mouse embryogenesis dataset (GSE117542) (2). Our reimplementation was conducted under limited computational resources, using 500 cells (down from the full dataset), a batch size of 32, and 500 training epochs (compared to 2000 in the original). Despite these simplifications, we reproduced key results: the learned embeddings showed lineage-consistent structure, and many top-ranked genes by tree-likeness overlapped with those from the original study.

We then experimented with two alternative distance metrics for the triplet loss. Manhattan distance preserved overall tree structure but yielded a slightly higher final loss. Cosine distance, in contrast, resulted in significantly higher triplet loss and more diffuse embeddings, likely due to its incompatibility with hierarchical relationships. These experiments confirmed the importance of distance metric selection in contrastive learning for lineage-based tasks.

To expand PORCELAN's applicability, we surveyed three additional datasets—GSE272484, GSE210172, and GSE200610—selected for their relevance to lineage tracing and regeneration biology. However, all three posed significant structural limitations, including the absence of explicit parent-child lineage trees, or inaccessible lineage metadata due to privacy concerns. While biologically rich, they were ultimately incompatible with PORCELAN's architecture, which depends on precise tree-distance information.

Lastly, we proposed theoretical improvements to PORCELAN's loss functions:

- A **depth-aware triplet margin** that adjusts contrastive penalties based on relative depth in the lineage tree.
- An **attention-weighted triplet loss** that emphasizes more reliable or biologically informative tree regions.
- A **smoothness regularization** term that enforces gradual change in the latent space across lineage paths.

Due to lack of GPU access and limited time in the latter part of the semester, we were unable to implement and test these enhancements. Nonetheless, our current implementation provides a robust baseline, and our proposed extensions offer a roadmap for improving lineage-aware representation learning in future work. This framework can play a critical role in emerging single-cell and spatial transcriptomics applications.

Supporting information (Optional)

Drive link to file (PDF) We have provided the following materials in our supplementary Google Drive folder:

`https://drive.google.com/drive/folders/1ZCU-iRY4777SWt9al_bCiMD-vMfk8RU-?usp=sharing`

- The project presentation slides (PDF)
- A supplementary PDF file containing all referenced images. These are referred to in the report as "Supp. Info. Fig. X"

All plots and figures referenced in the supplementary section (e.g., Supp. Info. Fig. 3) can be found in this PDF. These visualizations are not included in the main report due to the page limit.

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We would also like to thank the authors of the PORCELAN paper (1) and their GitHub repository (8) for making their codebase publicly available, which significantly helped our reimplementation and analysis. We are also grateful to the research community for sharing datasets on the GEO platform that allowed us to explore and evaluate lineage-traced biological data.

Author Contribution

Yash Mehta: Implemented and executed the PORCELAN model variants (Manhattan, Cosine), prepared datasets for processing, curated result visualizations, and compiled the supplementary materials. Also worked on formatting new datasets for compatibility with PORCELAN.

Aditya S: Focused on understanding and analyzing the core algorithmic components of PORCELAN, contributed to model reimplementation and proposed theoretical extensions to the triplet and autocorrelation losses.

Both authors collaborated equally in structuring the report, writing sections, interpreting results, and contributing to the final presentation.

GitHub Link

The project repository is publicly available at:

`https://github.com/YashMehta2094/CS6024_Project_Codes/tree/main`

Our code is adapted from the original PORCELAN tutorial (8), and can be run by replacing the `tutorial.ipynb` file with any of the provided versions in our repository (reimplementation, Manhattan distance, and Cosine distance).

Progress After Presentation

After our mid-semester presentation, our primary focus shifted to expanding PORCELAN's use beyond the original dataset. We identified and analyzed three biologically relevant datasets:

- **GSE272484**: A zebrafish liver regeneration dataset using the CellCousin system. Although it included single-cell expression data and metadata in '.h5ad' and '.csv' formats, it lacked an explicit parent-child lineage tree. The lineage structure was inferred only through injury status and developmental labels, which could not be mapped into the strict tree format required by PORCELAN.
- **GSE210172**: A mouse MERGE-seq dataset where barcoded neurons were traced to projection targets in the brain. While barcodes provided categorical groupings, they did not reflect developmental ancestry or tree distance, making them incompatible with PORCELAN's triplet and autocorrelation losses.
- **GSE200610**: This human dataset involved barcoded dopamine neuron grafts. However, only count matrices were publicly available, with raw lineage information withheld due to privacy concerns. This made it impossible to construct the required tree distances.

We invested significant effort in trying to adapt these datasets, including lineage tree approximation attempts and tree inference exploration, but ultimately concluded that structural incompatibility prevented their direct use in PORCELAN.

Alongside dataset exploration, we developed and documented several theoretical proposals for improving PORCELAN:

- **Depth-aware triplet margins**: Scale the triplet margin proportionally to tree depth, so more distant nodes are penalized more for being close in latent space.
- **Attention-weighted triplet loss**: Weight each triplet based on the confidence or biological informativeness of the anchor-positive-negative relationship. These weights could be manually defined or learned via backpropagation.
- **Smoothness regularization across tree paths**: Encourage gradual variation in the latent space by penalizing abrupt jumps between sequential nodes (e.g., parent to child to grandchild).

While we could not implement these ideas due to limited compute resources (lack of GPU access) and end-of-semester constraints, we believe they represent promising directions for future work in tree-structured representation learning. Our project's contributions—both empirical and theoretical—offer a strong foundation for others interested in extending PORCELAN or applying it to more complex developmental datasets.

References

1. Schlüter HM, Uhler C. Integrating representation learning, permutation, and optimization to detect lineage-related gene expression patterns. Nature Communications. 2025;16(1):1062.

2. Chan MM, Smith AZ, Grosswendt S, Kretzmer H, Norman T, Adamson B, et al.. CRISPR-based molecular recording of mouse embryogenesis; 2019. `https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE117542`.

3. Sema E, Singh SP. Cholangiocytes contribute to hepatocyte regeneration after partial liver injury in larval zebrafish; 2024. `https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE272484`.

4. Xu P, Peng J, Yuan T, Chen Z, Wu Z, Li CT, et al.. High-throughput mapping of single-neuron projection and molecular features by retrograde barcoded labelling [scRNA-seq]; 2024. `https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE210172`.

5. Storm P, Parmar M. Molecular barcodes and single cell transcriptomics to map graft diversity and lineage of human DA neurons in a rat xenograft model of Parkinson's Disease; 2023. `https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE200610`.

6. Jolliffe IT, Cadima J. Principal component analysis: a review and recent developments. Philosophical transactions of the royal society A: Mathematical, Physical and Engineering Sciences. 2016;374(2065):20150202.

7. McInnes L, Healy J, Melville J. Umap: Uniform manifold approximation and projection for dimension reduction. arXiv preprint arXiv:180203426. 2018;.

8. Lab CU. porcelan GitHub Repository; 2023. `https://github.com/uhlerlab/porcelan`.